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Stainless Steel - Grade 316

Grade 316 / 316L Technical Data

Summary

Grade 316 is an improved version of CS 304, with the addition of molybdenum and a slightly higher nickel content. The resultant composition of CS 316 gives the steel much increased corrosion resistance in many aggressive environments. The molybdenum makes the steel more resistant to pitting and crevice corrosion in chloride-contaminated media, sea water and acetic acid vapours. The lower rate of general corrosion in mildly corrosive environments gives the steel good atmospheric corrosion resistance in polluted marine atmospheres. CS 316 offers higher strength and better creep resistance at higher temperatures than CS 304. CS 316 also possesses excellent mechanical and corrosion properties at sub-zero temperatures. When there is a danger of corrosion in the heat-affected zones of weldments, the low-carbon variety CS 316L should be used. CS 316 Ti, the titanium-stabilised version, is used for its resistance to sensitization during prolonged exposure in the 550°C-800°C temperature range.

Typical Applications

Because of its superior corrosion and oxidation resistance, good mechanical properties and fabricability, CS 316 has applications in many sectors of industry. Some of these include:

Tanks and storage vessels for corrosive liquids.
Specialised process equipment in the chemical, food, paper, mining, pharmaceutical and petroleum industries.
Architectural applications in highly corrosive environments.

Chemical Composition (ASTM A 240)

	C	Mn	P	S	Si	Cr	Ni	Mo	Ti
SX316	0.08 max	2.0	0.045	0.030	1.0	16.0 to	10.0 to	2.00 to	0.5 max
SX316L	0.03 max	max	max	max	max	18.0	14.0	3.00	5X%C
SX316Ti	0.08 max								

Typical properties in the annealed condition

The properties quoted in this publication are typical of mill products and unless indicated must not be regarded as guaranteed minimum values for specification purposes.

1. Mechanical properties at room temperature

	SX316		SX316L		SX316Ti	
	Typical	Minimum	Typical	Minimum	Typical	Minimum
Tensile Strength, MPa	580	515	590	485	600	515
Proof Strength, (Offset 0.2%), MPa	310	205	310	170	320	205
Elongation (Percent in L = 5.65 S ₀)	55	40	60	40	50	40
Hardness (Brinell)	165	-	170	-	165	-
Erichsen Cup Test Value mm	8 - 10	-	240	-	-	-
Endurance (fatigue) limit, MPA	260	-	260	-	260	-

2. Properties at elevated temperatures

The values given refer to CS 316 and CS 316 Ti only as strength values for CS 316L fall rapidly above 425°C.

Short Time Elevated Temperature Tensile Strength

Temperature, °C	600	700	800	900	1000
Strength, MPa	460	320	190	120	70

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Creep data

Stress for a creep rate of 1% in 10 000 h

Temperature, °C	550	600	650	700	800
Stress, MPa	160	120	90	60	20

Recommended Maximum Service Temperature
(Oxidising conditions)

Continuous Service 925°C

Intermittent Service 870°C

3. Properties at Sub-Zero Temperatures (CS 316)

Temperature	°C	-78	-161	-196
Proof Strength (0.2% Offset)	MPa	400	460	580
Tensile Strength	MPa	820	1150	1300
Impact Strength (Charpy V-Notch)	J	120	165	155

4. Corrosion Resistance

4.1 Aqueous

For specific conditions, consult VRN technical staff. As a rough guide, the following examples are given for pure acid-water mixtures.

Temperature °C	20	80
Concentration, (-% by mass)	10 20 40 60 80 100	10 20 40 60 80 100
Sulphuric Acid	0 1 2 2 1 0	2 2 2 2 2 2
Nitric Acid	0 0 0 0 0 1	0 0 0 1 1 2
Phosphoric Acid	0 0 0 0 1 2	0 0 2 2 1 1
Formic Acid	0 0 0 1 1 0	0 2 2 2 2 0

Key: 0 = resistant - corrosion rate less than 100 >mm/year
1 = partly resistant - corrosion rate 100 m to 1000 >mm/year
2 = non resistant - corrosion rate more than 1000 mm/year

4.2 Atmospheric

The performance of CS 316 compared with other metals in various environments is shown in the following table. Corrosion rate is based on a 5 year exposure.

Environment	Corrosion Rate (µm/year)		
	SX 314	Aluminium-3S	Mild Steel
Rural	0.0025	0.025	5.8
Marine	0.0076	0.424	34.0
Marine Industrial	0.0051	0686	46.2

Note: For corrosion resistance of CS 316 relative to other types, see the section in Comparative Data.

http://askzn.co.za/draft2/tech/tech_grade_316.htm

4.3.1 Annealing. Heat from 1 010°C to 1 120°C and cool rapidly in air or water. The best corrosion resistance is obtained when the final annealing temperature is above 1 070°C.

4.3.2 Stress relieving. Heat from 200 - 400°C and air cool.

4.3.3 Hot working

Initial forging and pressing: 1150 - 1200°C

Finishing temperature: above 900°C

For upsetting operations, forgings

should be finished between: 930 and 980°C

All hot working operations should be followed by annealing.

Note: Soaking times to ensure uniformity of temperature are up to 12 times that required for the same thickness of mild steel.

Cold Working

CS 316 / 316L, being extremely tough and ductile, can be readily fabricated by cold working. Typical operations include bending, forming, deep drawing and upsetting.

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3CR12		Stainless Fabrication	Corrosion Institute
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		Cleaning Stainless Steel	Engineering News